

G-CAF

GovDigital Composable Architecture Framework

Volume 15

Platform Tier Deep Dive

v2.0

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Chapter 1

Developer Sub-Tier & IDP

The Developer sub-tier provides Internal Developer Platform (IDP) capabilities for self-service PBC development, deployment, and operations.

IDP Component	Capabilities
Developer Portal	Documentation, onboarding guides, support channels
Service Catalog	Discovery, metadata, dependency tracking
Golden Paths	Templates, scaffolds, blueprints for standard PBC patterns
CI/CD Pipelines	Build, test, deploy, release automation
Governance Automation	Policy as Code, compliance checks, audit trails
Composition Platform	UI module federation, assembly support

Chapter 2

Supply Chain Security

G-CAF adopts the SLSA (Supply-chain Levels for Software Artifacts) framework.

Threat Landscape

Stage	Threats	Real-World Examples
Source	Compromised developer, typosquat packages, malicious PRs, hijacking	npm (2021)
Build	Malicious build scripts, compromised CI/CD, dependency confusion	SolarWinds (2020), Codecov (2021)
Distribute	Registry poisoning, package hijacking, MITM, unsigned artifacts	Log4j (2021)

Chapter 3

SBOM & AIBOM Requirements

SBOM by Gateway

Gateway	SBOM Requirement
Initiation	SBOM generation planned in CI/CD
Design	SBOM format and storage defined
Go-Live	SBOM generated and stored (mandatory)
Retirement	SBOM archived with PBC artifacts

Supported formats: SPDX (preferred) and CycloneDX. Generated automatically by tools like Syft, Trivy, or language-specific generators.

AIBOM — AI Bill of Materials

Criticality	AIBOM Depth
Tier 1 (Critical)	Full AIBOM — all fields required (model identity, architecture, training data provenance, hyperparameters, dependencies)
Tier 2 (High)	Standard — core fields required, extended recommended
Tier 3 (Medium)	Basic — core identity and dependency fields
Tier 4 (Low)	Minimal — model identity and version only

AIBOM tracks: model name/version/ID, architecture type, framework, training data provenance, base model dependencies, known vulnerabilities, and performance characteristics.

Chapter 4

Container Signing

SAL Level	Signing	Key Management	Verification
SAL-1	Recommended	Keyless (OIDC)	Optional
SAL-2	Required	Keyless (OIDC)	Pre-deployment
SAL-3	Required	Keyless or HSM-backed	Enforced at admission
SAL-4	Required	HSM-backed	Enforced + attestations

Uses Sigstore/Cosign for keyless signing with transparency logs. Mitigates: registry compromise, build server compromise, MITM, image tag reassignment, insider threats, rollback attacks.

Chapter 5

Platform Foundation

The Bootstrap Problem

If all platform components were PBCs, circular dependencies would arise: the PBC Catalog PBC needs the database, the database needs Kubernetes, Kubernetes needs the IDP... Foundation components break this cycle.

The Core Question: Can this component be replaced with a different implementation while maintaining the same contracts? YES → Platform PBC (full gateway governance). NO → Foundation (IaC/GitOps governance).

Aspect	Platform Foundation	Platform PBC
Governance	IaC + GitOps (PR → Review → Merge → Argo CD)	G-CAF Gateway process
Replaceability	Single-instance; not replaceable	Can be swapped; contract-driven
Business Logic	None; infrastructure runtime	Contains domain/business logic
Ownership	Platform Team	Entity or central teams
Change Process	Standard/Significant IaC change	DDA gateway approval

Chapter 6

Foundation Components & IDP Foundation

Component	Category	Purpose	Consumers
AKS Kubernetes Cluster	Compute & Runtime	Container orchestration for all PBCs	All PBCs
PostgreSQL Cluster	Data Storage	Relational database for platform services	Catalog, Gateway, Registry PBCs
Vector DB	AI Storage	Embedding storage for semantic search	IDP Assistant, Agent PBCs
MCP Gateway	AI Integration	Model Context Protocol routing for tool use	Agent PBCs, IDP Assistant
LLM Gateway	AI Integration	LLM API routing, rate limiting, cost tracking	All AI-consuming PBCs
Backstage Shell	Developer Experience	IDP user interface runtime	Catalog, Golden Path, Scorecard PBCs

IDP Platform PBCs that run on the foundation: Catalog Service (PBC discovery), Golden Path Service (template management), Scorecard Service (compliance tracking), IDP Assistant (AI-powered developer help).

Chapter 8

Integration Sub-Tier & Data Fabric

The Integration sub-tier is the central nervous system for PBC communication and data unification.

Component	Sub-Tier	Function
API Gateway	Service Integration (GovDX)	API routing, rate limiting, authentication
Event Bus	Service Integration	Asynchronous PBC-to-PBC communication
Schema Registry	Service Integration	Contract version management, compatibility checks
Consumer Contracts	Service Integration	Consumer-specific API agreements
Data Virtualization	Data Integration (Fabric)	Unified data access without moving data
Active Metadata	Data Integration	Automated data cataloging and lineage
Knowledge Graphs	Data Integration	Entity relationships and semantic linking
Analytics Data Store	Data Integration	Medallion architecture (Bronze/Silver/Gold)
ML Training Data	Data Integration	Curated datasets for AI/ML training

Chapter 10

Infrastructure Sub-Tier

Digital Public Infrastructure (DPI)

DPI Component	Capabilities
Digital ID (UAE Pass)	Identity, authentication, federation
Payments (Pay.gov)	Collection, settlement, refunds
Base Registries	Population, business, geography registries
Digital Signatures	Signing, verification, archival
Notary Services	Timestamping, verification
Messaging Gateway	SMS, email, push notification delivery

Cloud Infrastructure includes sovereign cloud (Azure), Kubernetes platform, serverless compute, and managed services. This is the most stable sub-tier with the lowest rate of change.

Chapter 11

Environment Specification

Environment	Code	Purpose	Isolation	Data
Development	dev	Unit testing, local dev	Shared (many PBCs)	Synthetic only
Integration	int	Cross-PBC testing	Pooled (per team namespace)	Sanitized subset
Staging	stg	Pre-production validation	Tenant (per PBC isolated)	Production mirror
Pre-Production	ppd	Final validation	Dedicated (per PBC, HA)	Production mirror
Production	prd	Live service	Dedicated (per PBC, HA)	Real data

Self-service provisioning via IDP. Naming convention: {org}-{entity}-{pbc}-{env}. Environment lifecycle: Provisioning → Active → Scaling → Decommissioning.

Chapter 12

Domain Model Deep Dive

Bounded Context Classification

Classification	Investment	Sourcing	Example
Core	Highest — strategic advantage	Custom build	Citizen Identity, Digital Payments
Supporting	Medium — enables core	Build or configure	Case Management, Notifications
Generic	Lower — no differentiation	Buy or outsource	Email, Collaboration, HR

Context Map Patterns

Pattern	Description	G-CAF Usage
Upstream/Downstream	Producer influences consumer	Provider PBC → Consumer PBC
Customer/Supplier	Consumer drives producer priorities	Entity PBC → Platform PBC
Anti-Corruption Layer	Translation layer protecting model	Vendor PBC integration
Open Host Service	Well-defined protocol for any consumer	Standard contracts in GovDX
Published Language	Shared interchange format	CloudEvents, OpenAPI schemas
Shared Kernel	Shared model subset — AVOID in G-CAF	Creates NS; use Published Language instead

Aggregate Design Rules: single root entity, consistency boundary within single transaction, cross-aggregate references by ID only, prefer smaller aggregates, eventual consistency between aggregates. Domain Event naming: {context}.{aggregate}.{pastTenseVerb} (e.g., citizen.profile.registered). All events use CloudEvents specification.

Chapter 13

Analytics Experiences

Type	Audience	Delivery	Standards
Executive Dashboard	Leadership	Push (scheduled)	5-7 KPIs max, daily refresh, mobile required, <3s load
Operational Dashboard	Operations	Real-time	Live data, drill-down, alerting
Analytical Report	Analysts	Pull (on-demand)	Complex calculations, export support
Self-Service	Business users	Interactive	Ad-hoc exploration, discovery
Embedded Analytics	App users	In-app	Context-specific, lightweight
Paginated Report	Compliance	Scheduled	Document format, regulatory standard

Semantic Layer: 4-layer architecture — Table Layer (facts, dimensions, bridges) → Relationship Layer (joins, role-playing dims) → Calculation Layer (measures, KPIs) → Presentation Layer (folders, friendly names, descriptions).

Chapter 14

Four-Layer Governance Architecture

Layer	Code	Focus	Question Answered	G-CAF Implementation
Original	O	Governance decisions	What commitments are made?	16 Essential Decisions (GD-01–GD-16)
Information	I	Assessments & monitoring	What do we know?	40 Information Transactions (IT-01–IT-40)
Documentation	D	Contracts & records	What is recorded?	5 standard contracts + extensions
Material	M	Execution & operations	What is done?	16 Execution Transactions (MT-01–MT-16)

This COBIT-DEMO aligned model helps stakeholders understand the complete governance picture: decisions drive assessments, assessments inform documentation, documentation guides execution, and execution results feed back into information for the next decision cycle.